NAZARBAYEV UNIVERSITY

Graph-Based Visual Question Answering with Dynamic Graph Construction

CSCI 595 Generative AI

Aibek Abilgazym
Baimyrza Kalmyrzayev
Dina Almassova
Rustem Kaliyev

20.04.2026

NAZARBAYEV UNIVERSITY

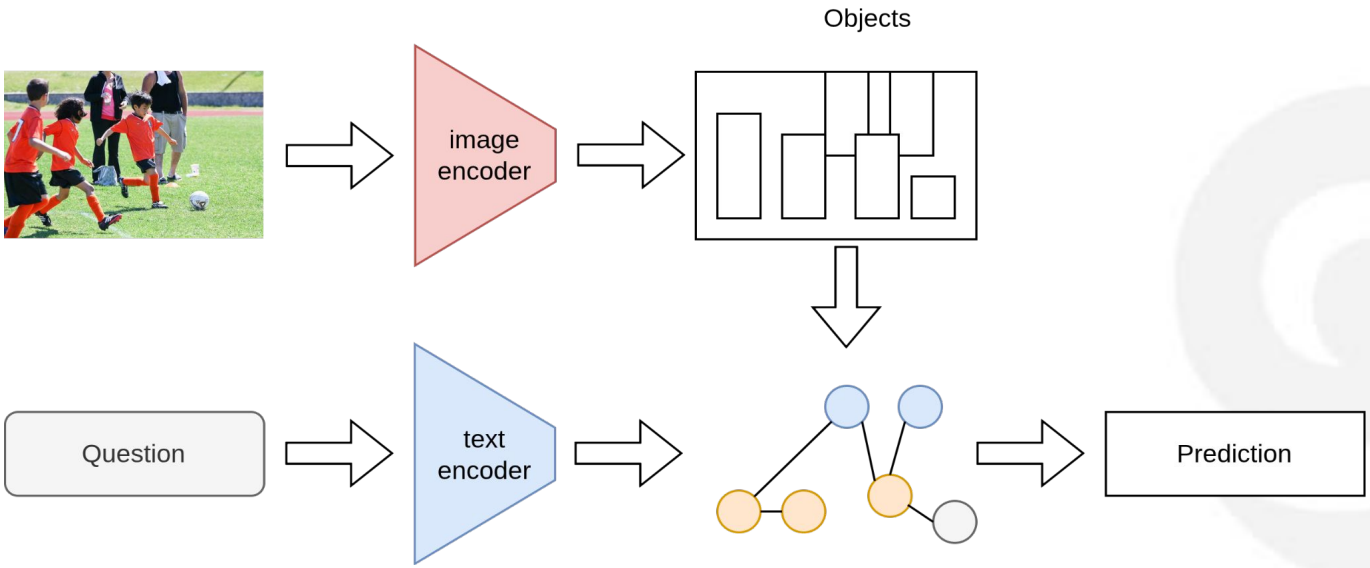
Outline

1. Introduction
2. Related Works
3. Challenges
4. Dataset
5. Proposed Methodology
6. Results

Introduction

Visual Question Answering (VQA) requires understanding both visual content and relationships between objects in an image in order to answer natural language questions.

- Graph structure may depend on the **text and/or image**
- Edges connect objects that are **relevant to the question**
- **Relational reasoning** on constructed graph



NAZARBAYEV UNIVERSITY

Related Work

Attention-based VQA

Bottom-Up and Top-Down Attention:

- Object-level features (Faster R-CNN)
- Top-down attention

MCAN:

- Deep modular co-attention (image-text)
- No explicit relational structure

Graph-based VQA

ReGAT

- Graph attention over visual regions
- Static and image-conditioned graph

GraphVQA

- Language-guided GNN
- Static graph from annotated scene graphs

Dynamic Graphs

DGCN

- Dynamic graph construction
- Learned graph structure

Our Motivation

NAZARBAYEV UNIVERSITY

Challenges

Codebase Setup & Dependencies

- GraphVQA codebase is outdated ⇒ harder to reproduce
- Significant debugging of packages (PyTorch Geometric)

Running the Baseline

- GraphVQA codebase contains a lot of commented code and undocumented steps (3 metrics were commented)

Extending the Codebase

- Integrating our idea required modifying the core message-passing logic without breaking the existing pipeline

NAZARBAYEV UNIVERSITY

The GQA Dataset (1/2)

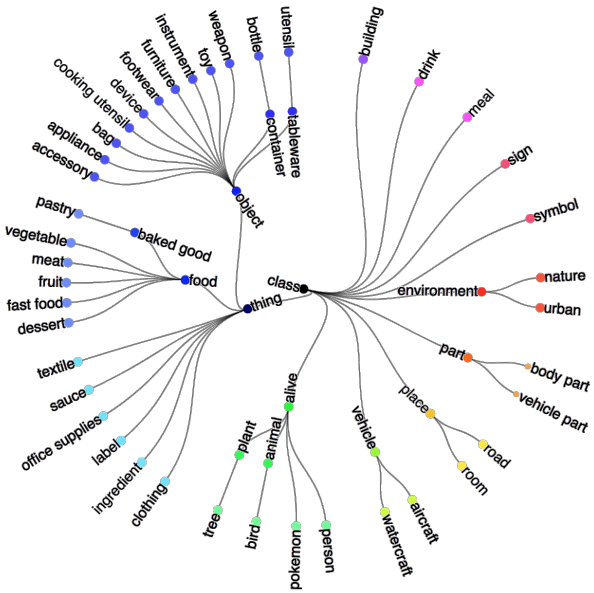
https://cs.stanford.edu/people/dorarad/gqa/

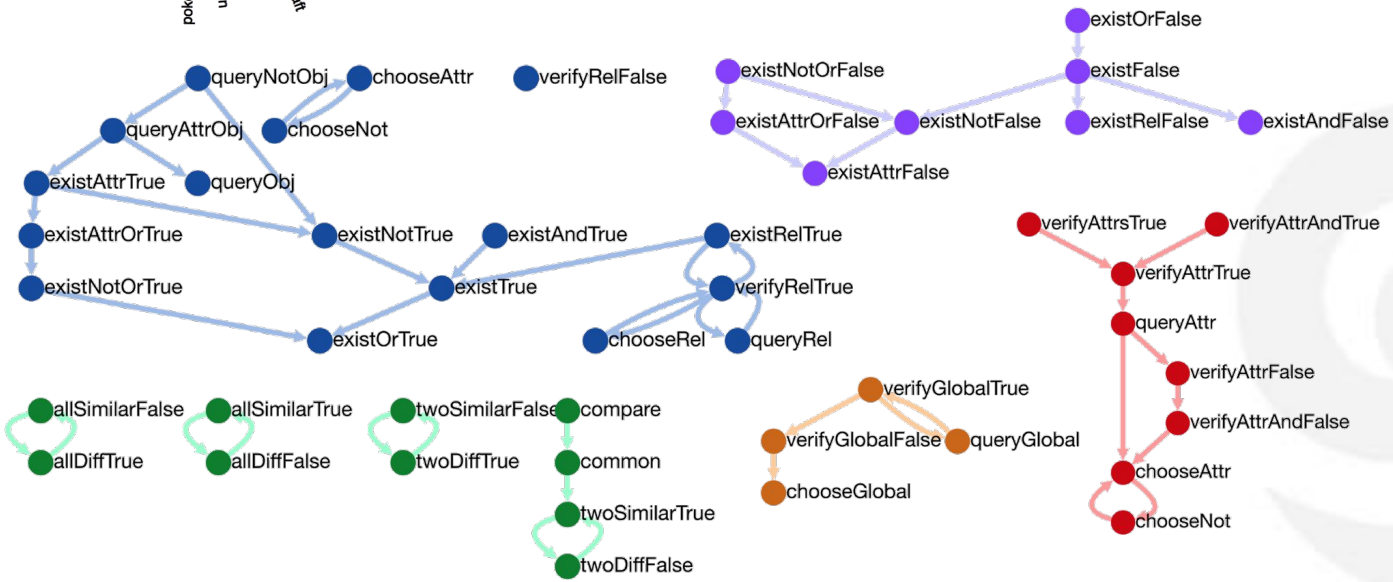
Size: 22M multi-step questions with 113,018 real-world images
Every question is paired with a functional program and is grounded in Visual Genome scene graphs representing objects, attributes, and relations

Input:

- Image: Standard RGB tensors (3 x 224 x 224) or pre-extracted object bounding box features (100 x 2048 dimensional vectors per image, 114 GB) and spatial location features (56 GB).
- Text: Tokenized natural language questions.
- Metadata: Scene graphs detailing objects, attributes, and spatial relations.

The GQA Dataset (2/2)





NAZARBAYEV UNIVERSITY

Proposed Methodology (1/4)

Existing implementation (GraphVQA)

- Objects and their relationships are taken directly from GQA scene graph annotations
- Question is parsed into instruction vectors using Seq2Seq transformer, which condition the message passing at each GNN layer by concatenation
- Different GNNs (GAT, GCN, GINE) are used as message passing models

Limitations

1. The graph structure is static, the same for every question asked about the same image
2. Scene graph annotations can be incomplete
3. All annotated edges participate in message passing even if they are irrelevant to the asked question

NAZARBAYEV UNIVERSITY

Proposed Methodology (1/4)

Existing implementation (GraphVQA)

- Objects and their relationships are taken directly from GQA scene graph annotations
- Question is parsed into instruction vectors using Seq2Seq transformer, which condition the message passing at each GNN layer by concatenation
- Different GNNs (GAT, GCN, GINE) are used as message passing models

Limitations

1. The graph structure is static, the same for every question asked about the same image
2. Scene graph annotations can be incomplete
3. All annotated edges participate in message passing even if they are irrelevant to the asked question

NAZARBAYEV UNIVERSITY

Proposed Methodology (2/4)

Dynamic Graph Construction

We propose three formulations of question-conditioned graph construction, each representing a different inductive bias

Approach 1: Edge reweighting (**Limitation 1**)

- Keep all edges, then reweight them with question-conditioned gate
- ***Hypothesis***: not all annotated edges are equally relevant to every question

$$e_{ij} = \mathrm{softmax}(\sigma(\mathrm{MLP}(h_i, h_j, q)) \odot \mathrm{MLP}(h_i, h_j, e_{ij}))$$

NAZARBAYEV UNIVERSITY

Proposed Methodology (3/4)

Approach 2: Edge augmentation (**Limitation 2**)

- Keep original scene graph edges, dynamically add top-k missing connections
- ***Hypothesis***: scene graphs are incomplete, question can recover missing edges

$$\mathcal{E}_{\text{aug}} = \mathcal{E}_{\text{static}} \cup \text{top-k}(\text{MLP}(h_i, h_j, q))$$

Approach 3: Edge pruning (**Limitation 3**)

- Retain only top-k most question-relevant edges and remove the rest
- ***Hypothesis***: focused sparse graphs improve interpretability and reduce noise

$$\mathcal{N}_i^{\text{pruned}} = \text{top-k}_{j \in \mathcal{N}_i}(\text{MLP}(h_i, h_j, q))$$

Quantitative metric: Accuracy on GQA evaluation set

Qualitative metric: Visualization of edge maps (different graph structures for the same image with different questions)

NAZARBAYEV UNIVERSITY

Proposed Methodology (4/4)

Metrics

Program – token-level exact match accuracy over individual program instructions. A single instruction is counted correct only if every token matches exactly

Program Group – stricter version of **Program**. A question's entire program must be fully correct to count as a match. One wrong instruction fails the whole question

Program Non Empty – same as **Program** but excludes empty instructions (padding-only)

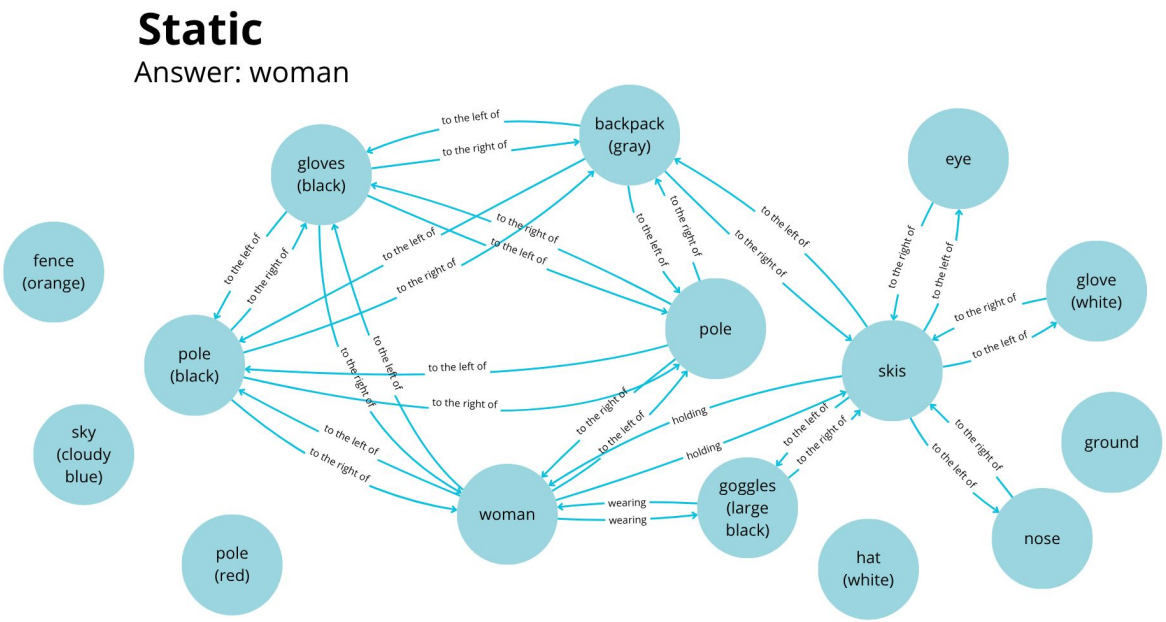
Short – classification accuracy on the final short answer ("yes", "red", etc.). Top-1 classification over all classes.

NAZARBAYEV UNIVERSITY

Results (1/5)

Model	Program	Program Group	Program Non Empty	Short	Train Time (mins/epoch)
Static	96.75	87.25	94.72	90.13	6.61
Reweight	96.72	87.15	94.67	91.51	6.8
Augment	96.65	87.08	94.56	93.51	38.47
Prune	96.65	87.11	94.57	93.50	18.61

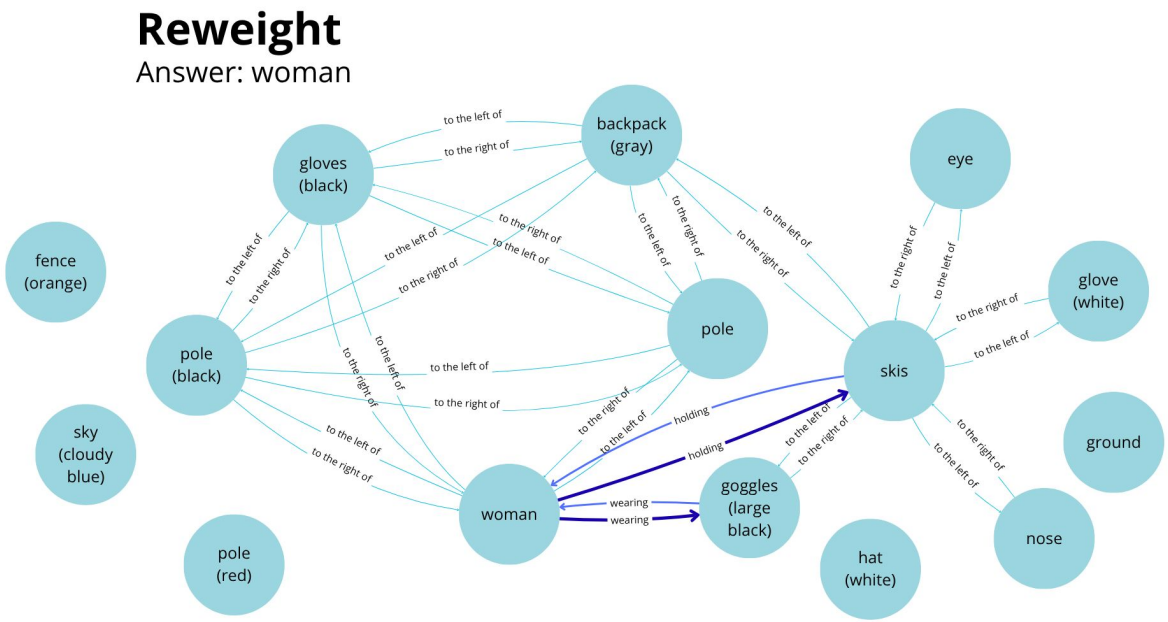
Results (2/5)



Q:Who is wearing goggles?

A: Woman

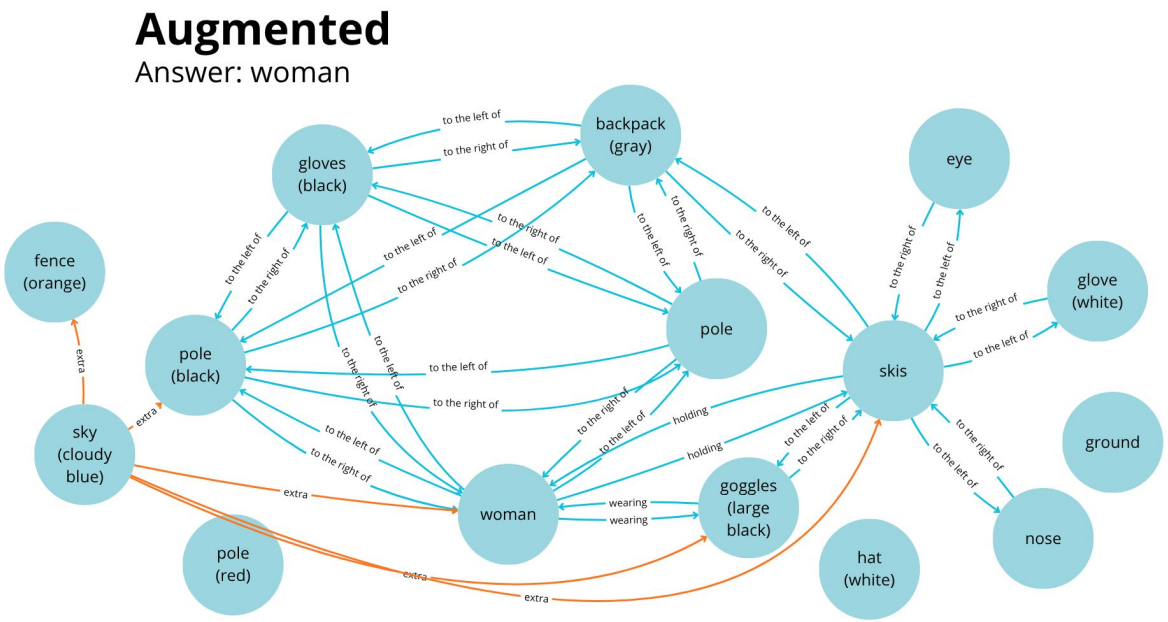
Results (3/5)



Q:Who is wearing goggles?

A: Woman

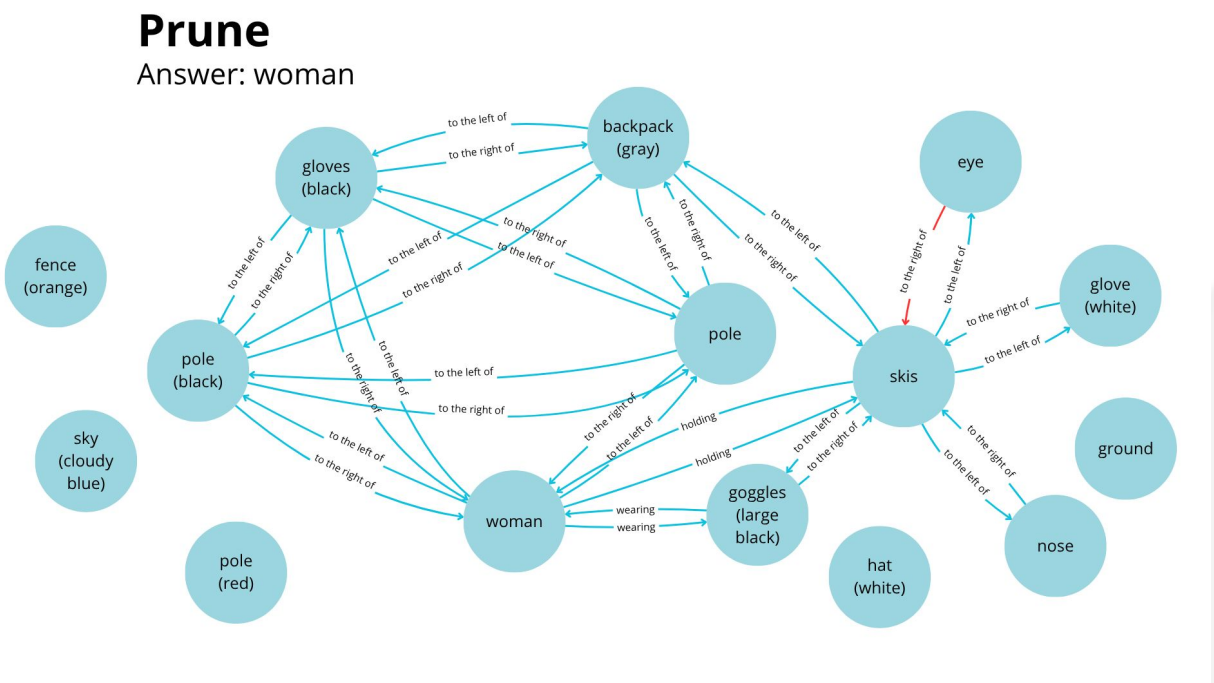
Results (4/5)



Q:Who is wearing goggles?

A: Woman

Results (5/5)



Q:Who is wearing goggles?

A: Woman

NAZARBAYEV UNIVERSITY

References

[1] P. Anderson, X. He, C. Buehler, D. Teney, M. Johnson, S. Gould, and L. Zhang, "Bottom-up and top-down attention for image captioning and visual question answering," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6077–6086, 2018.

[2] Z. Yu, J. Yu, Y. Cui, D. Tao and Q. Tian, "Deep modular co-attention networks for visual question answering," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6281–6290, 2019.

[3] L. Li, Z. Gan, Y. Cheng, and J. Liu, "Relation-aware graph attention network for visual question answering," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10313–10322, 2019.

[4] W. Liange, Y. Jiang, and Z. Liu, "Graphvqa: Language-guided graph neural networks for scene graph question answering," in *NAACL*, 2021

[5] Z. Qiu, K. Qiu, J. Fu, and D. Fu, "Dgcn: Dynamic graph convolutional network for efficient multi-person pose estimation," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 07, pp. 11924–11931, 2020.